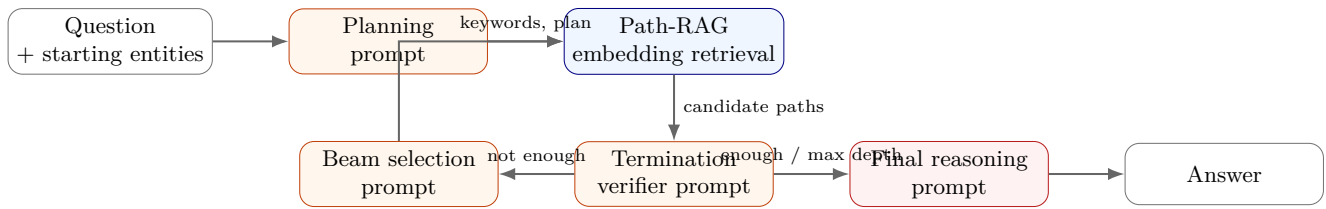


FiDeLiS Internal Prompt Map

Executive Finding

The current FiDeLiS pipeline is not strictly KG-faithful at the final answer step. For WebQSP, CWQ, and CR-LT, the final answer prompt allows the model to answer based on the retrieved reasoning path *and your knowledge*. This makes it possible for correct answers to come from the LLM's pretrained parametric knowledge rather than from the retrieved KG path.

Pipeline View



Runtime Message Assembly

Every chat prompt is assembled in `src/utils/llm_backbone.py` as:

Message format

[system prompt] → [few-shot examples] → [user prompt template]. The current generation parameters are temperature=0, top_p=0, and logprobs=False.

Prompt Inventory

Pipeline stage	Prompt object	Role in pipeline	Risk
Planning / keyword generation	<code>plan_prompt</code>	Produces keywords, planning steps, and declarative statement.	Medium
Path-RAG retrieval	No chat prompt	Embeds LLM-generated keywords and ranks KG relations/neighbors.	Medium
Path verbalization	<code>reasoning_path_parser_prompt</code>	Converts symbolic KG path into natural language.	Medium
Termination verifier	<code>terminals_prune_single_prompt</code> or <code>deductive_verifier_prompt</code>	Decides whether a path is sufficient or deductively supports a statement.	High
Beam candidate selection	<code>beam_search_prompt</code>	Chooses top candidate path indices.	Medium
Final answer	<code>reasoning_prompt</code>	Produces the final answer from selected paths.	Critical

Dataset flag	Prompt module	Final-answer risk phrase	Risk
RoG-webqsp	src/prompts/webqsp.py	and your knowledge	Critical
RoG-cwq	src/prompts/cwq.py	and your knowledge	Critical
CL-LT-KGQA	src/prompts/cl_lt_kgqa.py	and your knowledge	Critical

Dataset Prompt Modules

Faithfulness Risk Points

- **Final answer can use pretrained knowledge.** The prompt explicitly permits parametric knowledge, so high accuracy does not prove KG-grounded reasoning.
- **CR-LT True/False is vulnerable to random guessing and prior knowledge.** A binary task can achieve about 50% by guessing, and the current prompt does not require KG-only entailment.
- **The verifier is LLM-mediated.** The verifier can judge path sufficiency using outside knowledge unless the prompt forbids it.
- **Path verbalization can add information.** The parser may transform a sparse symbolic path into a stronger natural-language claim than the KG path directly supports.

Representative Prompt Snippets

Final answer prompt, WebQSP/CWQ

Given a question and the associated retrieved reasoning path from a knowledge graph, you are asked to answer the following question based on the reasoning path and your knowledge.
 Question: {question}
 Reasoning path: {reasoning_path}
 Only return the answer to the question.

Final answer prompt, CR-LT

The CR-LT system message asks for True or False, but the final user prompt still uses the same wording: *based on the reasoning path and your knowledge*.

Deductive verifier prompt

Whether the conclusion {declarative_statement} can be deduced from {parsed_reasoning_path}, if yes, return yes, if no, return no.

Pointers

The full prompt dump with all examples is available in:

- docs/fidelis_pipeline_prompts.md
- docs/fidelis_pipeline_prompts.txt